Linear Transformations II

Jin-Ren Ryan Wang

July 31, 2026

Contents

1 Isomorphism of vector spaces

1.1 Definition

Definition 5.1.1. (inverse function) A function $f : A \to B$ is called *bijective* if f is both injective and surjective. If $f : A \to B$ is bijective, then its inverse function $f^{-1} : B \to A$ exists and satisfies $f^{-1}(f(a)) = a, \quad \forall a \in A$ and $f(f^{-1}(b)) = b, \quad \forall b \in B$.

Definition 5.1.2. Let V and W be vector spaces over a field F.

1. $f : V \to W$ is called a *(linear) isomorphism from V onto W* if f is a bijective linear map.

2. We say that V and W are *isomorphic as vector spaces* if there exists a linear isomorphism $f : V \to W$.

1.2 Theorem

Theorem 5.1.1. Let $f : V \to W$ be a linear map of vector spaces. If f is bijective (and hence f^{-1} exists), then $f^{-1} : W \to V$ is also linear.

Proof. Since f is surjective, every $\mathbf{w} \in W$ can be written as $\mathbf{w} = f(\mathbf{v})$ for some $\mathbf{v} \in V$. We verify the two properties of linearity.

1. **Additivity.**

   Let $\mathbf{w}_1 = f(\mathbf{v}_1), \mathbf{w}_2 = f(\mathbf{v}_2)$, where $\mathbf{v}_1, \mathbf{v}_2 \in V$. Then

$$\begin{aligned} f^{-1}(\mathbf{w}_1 + \mathbf{w}_2) &= f^{-1}(f(\mathbf{v}_1) + f(\mathbf{v}_2)) \\ &= f^{-1}(f(\mathbf{v}_1 + \mathbf{v}_2)) \\ &= \mathbf{v}_1 + \mathbf{v}_2 \\ &= f^{-1}(\mathbf{w}_1) + f^{-1}(\mathbf{w}_2). \end{aligned}$$

2. **Homogeneity.**

   Let $\lambda \in F$, $\mathbf{w} \in W$, so $\mathbf{w} = f(\mathbf{v})$ for some $\mathbf{v} \in V$. Then

$$\begin{aligned} f^{-1}(\lambda \mathbf{w}) &= f^{-1}(\lambda f(\mathbf{v})) \\ &= f^{-1}(f(\lambda \mathbf{v})) \\ &= \lambda \mathbf{v} \\ &= \lambda f^{-1}(\mathbf{w}). \end{aligned}$$

This shows $f^{-1} : W \to V$ is also linear. $\square$

Theorem 5.1.2. Let V be a vector space over a field F of dimension n. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis of V. Then the function $\phi_S : V \to F^n$ defined by

$$\mathbf{v} \mapsto [\mathbf{v}]_S.$$

is a linear isomorphism.

Proof. Write

$$\mathbf{v} = \sum_{i=1}^{n} \alpha_i \mathbf{v}_i, \qquad \mathbf{v}' = \sum_{i=1}^{n} \alpha_i' \mathbf{v}_i.$$

We prove the statement in three steps.

1. $\boxed{\phi_S \text{ is linear.}}$

   By definition, $[\mathbf{v}]_S = (\alpha_1, \alpha_2, \dots, \alpha_n)$.

   Hence

$$\begin{aligned}
\phi_S(\mathbf{v} + \mathbf{v}') &= \phi_S\left(\sum_{i=1}^{n} (\alpha_i + \alpha_i')\mathbf{v}_i\right) \\
&= (\alpha_1 + \alpha_1', \dots, \alpha_n + \alpha_n') \\
&= (\alpha_1, \dots, \alpha_n) + (\alpha_1', \dots, \alpha_n') \\
&= \phi_S(\mathbf{v}) + \phi_S(\mathbf{v}').
\end{aligned}$$

   Likewise, for any $\lambda \in F$,

$$\begin{aligned}
\phi_S(\lambda \mathbf{v}) &= \phi_S\left(\sum_{i=1}^{n} (\lambda \alpha_i)\mathbf{v}_i\right) \\
&= (\lambda \alpha_1, \dots, \lambda \alpha_n) \\
&= \lambda(\alpha_1, \dots, \alpha_n) \\
&= \lambda \phi_S(\mathbf{v}).
\end{aligned}$$

   Therefore, ϕ_S is linear.

2. $\boxed{\phi_S \text{ is injective.}}$

   Let $\mathbf{v} \in \ker(\phi_S) \implies \phi_S(\mathbf{v}) = (0, \dots, 0). \implies \mathbf{v} = \sum_{i=1}^{n} 0\, \mathbf{v}_i = \mathbf{0}_V$.

   Thus

$$\ker(\phi_S) = \{\mathbf{0}\},$$

   and therefore ϕ_S is injective.

3. $\boxed{\phi_S \text{ is surjective.}}$

   By the Rank–Nullity Theorem,

$$\text{rank}(\phi_S) + \text{nullity}(\phi_S) = \underbrace{\dim_F(V)}_{\text{n}}.$$

   Since ϕ_S is injective,

$$\text{nullity}(\phi_S) = 0.$$

   Hence

$$\text{rank}(\phi_S) = n = \dim_F(V).$$

   Because $\dim_F(F^n) = n$, we conclude that $\text{im}(\phi_S) = F^n$. Therefore, ϕ_S is surjective.

Hence ϕ_S is a linear isomorphism. □

Theorem 5.1.3. Let V and W be two finite-dimensional vector spaces over a field F. Then$V$ and W are isomorphic as vector spaces if and only if $\dim_F(V) = \dim_F(W)$.

Proof. We prove both directions.

($\Rightarrow$) Suppose V and W are isomorphic. By definition, $\exists$ a linear isomorphism $f : V \to W$

Since f is injective $\iff$ nullity$(f) = 0$, and f is surjective $\iff$ rank$(f) = \dim_F(W)$.

By the Rank–Nullity Theorem, rank(f) + nullity$(f) = \dim_F(V)$.

Hence $\dim_F(W) + 0 = \dim_F(V)$, so $\dim_F(V) = \dim_F(W)$.

($\Leftarrow$) Suppose $\dim_F(V) = \dim_F(W) = n$. Fix a basis S of V, and a basis S' of W.

By Theorem 5.1.2, the coordinate maps

$$\phi_S : V \to F^n, \qquad \phi_{S'} : W \to F^n$$

are linear isomorphisms.

Now, $V \xrightarrow{\phi_S} F^n \xrightarrow{\phi_{S'}^{-1}} W$

Therefore, their composition $\phi_{S'}^{-1} \circ \phi_S : V \to W$ is also a linear isomorphism.

Hence, $V \cong W$.

□

1.3 Example

1.3.1 Trivial Example

Let V be a vector space and let $\mathrm{id}_V : V \to V$ be the identity map on V. Then

Solution. Recall

$$\mathrm{id}_V(\mathbf{v}) = \mathbf{v}, \quad \forall\, \mathbf{v} \in V.$$

So

1. id_V is a linear map.
2. $\ker(\mathrm{id}_V) = \{\mathbf{0_V}\}$, hence id_V is injective.
3. $\mathrm{im}(\mathrm{id}_V) = \underbrace{V}_{\text{entire domain}}$, hence id_V is surjective.

Therefore, $\mathrm{id}_V : V \to V$ is a linear isomorphism. Consequently, V is isomorphic to itself. Also, $\boxed{\text{Every vector space is isomorphic to itself}}$

□

2 Representations of Linear Maps as Matrices

In this section, we describe a systematic way to represent every linear map
$f : V \to W$ by a concrete matrix.
To do this, we begin a linear map $f : V \to W$ where $\dim_F(V) = n$ and $\dim_F(W) = m$.
Suppose we have fixed

$$B_1 = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\} \text{ a basis of } V$$

$$B_2 = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_m\} \text{ a basis of } W$$

Since f is linear, it is completely determined by its values on the basis vectors of V. For each $\mathbf{v}_j \in B_1$, we can write

$$f(\mathbf{v}_j) = \alpha_{1j}\mathbf{w}_1 + \alpha_{2j}\mathbf{w}_2 + \cdots + \alpha_{mj}\mathbf{w}_m, \qquad j = 1, \ldots, n.$$

That is, we express each $f(\mathbf{v}_j)$ as a linear combination of the basis vectors of W.

We now collect the coefficients into a matrix. Specifically, we define

$$[f]_{B_1}^{B_2} = (\alpha_{ij})_{i,j} = \begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{m1} & \alpha_{m2} & \cdots & \alpha_{mn} \end{pmatrix},$$

whose j-th column is exactly the coordinate vector of $f(\mathbf{v}_j)$ with respect to the basis B_2.

2.1 Definition

Definition 5.2.1. The matrix constructed above is called the *matrix representing f with respect to the bases B_1 and B_2*. We denote it by $[f]_{B_1}^{B_2}$.

Definition 5.2.2. Let $A = (\alpha_{ij})$ be an $l \times m$ matrix and $B = \beta_{ij}$ be an $m \times n$ matrix. Then the product AB is a $l \times n$ matrix $C = (\gamma_{ij})$ where,

$$(i,j)\text{-entry of } AB = \gamma_{ij} = \sum_{k=1}^{m} \alpha_{ik}\beta_{kj}$$

2.2 Theorem

Theorem 5.2.1. Let $f : V \to W$ be a linear map between finite-dimensional vector spaces over a field F. Fix a basis B_1 be a basis of V, and B_2 be a basis of W.
Then, for every $\mathbf{v} \in V$, we have the equality of matrices (and vectors):

$$[f]_{B_1}^{B_2}[\mathbf{v}]_{B_1} = [f(\mathbf{v})]_{B_2}.$$

Proof. Let $B_1 = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, $B_2 = \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$
Let

$$[f]_{B_1}^{B_2} = (\alpha_{ij})_{i,j} \iff f(\mathbf{v}_j) = \sum_{i=1}^{m} \alpha_{ij}\mathbf{w}_i, \forall j$$

And let

$$[\mathbf{v}]_{B_1} = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} \iff \mathbf{v} = \sum_{j=1}^{n} \lambda_j \mathbf{v}_j$$

Then

$$[f]_{B_1}^{B_2}[\mathbf{v}]_{B_1} = (\alpha_{ij})_{i,j} \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^{n} \alpha_{1j}\lambda_j \\ \vdots \\ \sum_{j=1}^{n} \alpha_{mj}\lambda_j \end{pmatrix}.$$

On the other hand, since f is linear,

$$f(\mathbf{v}) = f\left(\sum_{j=1}^{n} \lambda_j \mathbf{v}_j\right) = \sum_{j=1}^{n} \lambda_j f(\mathbf{v}_j) = \sum_{j=1}^{n} \lambda_j \left(\sum_{i=1}^{m} \alpha_{ij}\mathbf{w}_i\right) = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} \alpha_{ij}\lambda_j\right) \mathbf{w}_i.$$

Hence,

$$[f(\mathbf{v})]_{B_2} = \begin{pmatrix} \sum_{j=1}^{n} \alpha_{1j}\lambda_j \\ \vdots \\ \sum_{j=1}^{n} \alpha_{mj}\lambda_j \end{pmatrix},$$

which is exactly the vector obtained above. Therefore,

$$[f]_{B_1}^{B_2}[\mathbf{v}]_{B_1} = [f(\mathbf{v})]_{B_2}.$$

Remark. The theorem can be summarized by the following commutative diagram:

$$\begin{array}{ccc} V & \xrightarrow{f} & W \\ \downarrow \phi_{B_1} & & \downarrow \phi_{B_2} \\ F^n & \xrightarrow{\color{red}{[f]_{B_1}^{B_2}}} & F^m \end{array}$$

That is,

$$\color{red}{[f]_{B_1}^{B_2}} \circ \phi_{B_1} = \phi_{B_2} \circ f,$$

or equivalently,

$$\color{red}{[f]_{B_1}^{B_2}}[\mathbf{v}]_{B_1} = [f(\mathbf{v})]_{B_2}, \qquad \forall\, \mathbf{v} \in V.$$

□

□

2.3 Example

2.3.1 Matrix Representation of Differentiation

Consider the linear map $D : \mathbb{R}[x]_{\leq 3} \to \mathbb{R}[x]_{\leq 2}$ defined by $p(x) \mapsto p'(x)$. Fix the bases

$$S_1 = \{1, x, x^2, x^3\} \text{ of } \mathbb{R}[x]_{\leq 3}, \text{ and } S_2 = \{1, x, x^2\} \text{ of } \mathbb{R}[x]_{\leq 2}.$$

Find the matrix $[D]_{S_1}^{S_2}$

Solution. We compute the images of the basis vectors of S_1
and express them in terms of the basis S_2.

First,

$$D(\color{red}{1}) = 0 = \color{red}{0} \cdot 1 + \color{red}{0} \cdot x + \color{red}{0} \cdot x^2,$$

so

$$[D(1)]_{S_2} = (0, 0, 0)^T.$$

Next,

$$D(\color{red}{x}) = 1 = \color{red}{1} \cdot 1 + \color{red}{0} \cdot x + \color{red}{0} \cdot x^2,$$

so

$$[D(x)]_{S_2} = (1, 0, 0)^T.$$

Similarly,

$$D(x^2) = 2x = 0 \cdot 1 + 2 \cdot x + 0 \cdot x^2, \quad [D(x^2)]_{S_2} = (0, 2, 0)^T,$$

and

$$D(x^3) = 3x^2 = 0 \cdot 1 + 0 \cdot x + 3 \cdot x^2, \quad [D(x^3)]_{S_2} = (0, 0, 3)^T.$$

Placing these coordinate vectors as columns, we obtain

$$[D]_{S_1}^{S_2} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

Remark. For example, if

$$p(x) = 1 + 2x + 4x^3, \qquad D[p(x)] = 2 + 12x^2$$

then

$$[p(x)]_{S_1} = \begin{pmatrix} 1 \\ 2 \\ 0 \\ 4 \end{pmatrix}, \qquad [D(p(x))]_{S_2} = \begin{pmatrix} 2 \\ 0 \\ 12 \end{pmatrix}.$$

We observe that

$$[D(p(x))]_{S_2} = [D]_{S_1}^{S_2}[p(x)]_{S_1}.$$

Thus differentiation has been translated into matrix multiplication. □

□

2.3.2 Matrix Representations under Different Bases

Consider the linear map $f : \mathbb{R}^3 \to \mathbb{R}^3$, defined by

$$(x, y, z) \longmapsto (-x - 2y + 2z,\ 4x + 3y - 4z,\ -2y + z).$$

(a) Verify that f is a linear transformation.

(b) Find the matrix representing f with respect to the standard basis of $\mathbb{R}^3$ (for both the domain and codomain).

(c) Verify that

$$S = \{(1, 0, 1), (0, 1, 1), (1, -1, 1)\}$$

is a basis of $\mathbb{R}^3$. Find the matrix $[f]_S^S$.

Solution. The following are the solutions of three subproblems

(a)

For any $\mathbf{u} = (x_1, y_1, z_1)$, $\mathbf{v} = (x_2, y_2, z_2) \in \mathbb{R}^3$,

$$\begin{aligned} f(\mathbf{u}+\mathbf{v}) &= f(x_1+x_2,\, y_1+y_2,\, z_1+z_2) \\ &= \Big(-(x_1+x_2) - 2(y_1+y_2) + 2(z_1+z_2), \\ &\qquad 4(x_1+x_2) + 3(y_1+y_2) - 4(z_1+z_2), \\ &\qquad -2(y_1+y_2) + (z_1+z_2)\Big) \\ &= f(\mathbf{u}) + f(\mathbf{v}). \end{aligned}$$

Moreover, for any $\lambda \in \mathbb{R}$,

$$\begin{aligned} f(\lambda\mathbf{u}) &= f(\lambda x_1, \lambda y_1, \lambda z_1) \\ &= \Big(-\lambda x_1 - 2\lambda y_1 + 2\lambda z_1,\, 4\lambda x_1 + 3\lambda y_1 - 4\lambda z_1,\, -2\lambda y_1 + \lambda z_1\Big) \\ &= \lambda f(\mathbf{u}). \end{aligned}$$

Therefore, f satisfies both additivity and homogeneity. Hence, $f : \mathbb{R}^3 \to \mathbb{R}^3$ is a linear transformation.

(b) Let $B = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ be the standard basis of $\mathbb{R}^3$.

Compute the images of the basis vectors:

$$f(\mathbf{e}_1) = (-1, 4, 0) = (-1) \quad \mathbf{e}_1 + 4 \quad \mathbf{e}_2 + 0 \quad \mathbf{e}_3,$$

$$f(\mathbf{e}_2) = (-2, 3, -2) = (-2) \quad \mathbf{e}_1 + 3 \quad \mathbf{e}_2 + (-2) \quad \mathbf{e}_3,$$

$$f(\mathbf{e}_3) = (2, -4, 1) = 1 \quad \mathbf{e}_1 + -4 \quad \mathbf{e}_2 + 1 \quad \mathbf{e}_3.$$

Hence,

$$[f]_B^B = \begin{pmatrix} -1 & -2 & 2 \\ 4 & 3 & -4 \\ 0 & -2 & 1 \end{pmatrix}.$$

(c)

Let

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{(1,0,1), (0,1,1), (1,-1,1)\}.$$

Compute

$$\begin{aligned} f(\mathbf{v}_1) &= (1,0,1) = 1\mathbf{v}_1 + 0\mathbf{v}_2 + 0\mathbf{v}_3, \\ f(\mathbf{v}_2) &= (0,-1,-1) = 0\mathbf{v}_1 + (-1)\mathbf{v}_2 + 0\mathbf{v}_3, \\ f(\mathbf{v}_3) &= (3,-3,3) = 0\mathbf{v}_1 + 0\mathbf{v}_2 + 3\mathbf{v}_3. \end{aligned}$$

Therefore,

$$[f]_S^S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Remark. Although the moral standard basis is convenient for computation, it is not necessarily the "nicest" basis for representing a linear transformation.
Question: How to find the "nice" basis for a linear map f?
Answer: Diagonalization Theory □

□

2.3.3 Using Matrix Representations to Study a Linear Map

Consider the function $f : \mathbb{R}[x]_{\le 2} \longrightarrow \mathbb{R}[x]_{\le 3}$

$$p(x) \longmapsto -\frac{d}{dx}\big(p(x^2-1)\big) + \int_2^x p(t)\,dt.$$

(a) Verify that f is a linear transformation.

(b) Find $\ker(f)$. Is f injective?

(c) Find $\operatorname{rank}(f)$. Is f surjective?

Solution. The following are the solutions of abobe three subproblems

(a)
Let $p(x), q(x) \in \mathbb{R}[x]_{\le 2}$ and let $\lambda \in \mathbb{R}$.
Observe that differentiation, composition with x^2-1, and integration are all linear operators. Hence

$$\begin{aligned} f(p+q) &= -\frac{d}{dx}\big((p+q)(x^2-1)\big) + \int_2^x (p(t)+q(t))\,dt \\ &= -\frac{d}{dx}\big(p(x^2-1)\big) - \frac{d}{dx}\big(q(x^2-1)\big) \\ &\quad + \int_2^x p(t)\,dt + \int_2^x q(t)\,dt \\ &= f(p) + f(q). \end{aligned}$$

Similarly,

$$f(\lambda p) = -\frac{d}{dx}\big(\lambda p(x^2-1)\big) + \int_2^x \lambda p(t)\,dt = \lambda\left(-\frac{d}{dx}(p(x^2-1)) + \int_2^x p(t)\,dt\right) = \lambda f(p).$$

Therefore, f satisfies additivity and homogeneity. Hence, f is a linear transformation.

(b)
Fix a basis $S_1 = \{1, x, x^2\}$ (domain), and a basis $S_2 = \{1, x, x^2, x^3\}$ (co-domain).
Compute the images of the basis vectors:

$$f(1) = x - 2 = (-2) + 1x + 0x^2 + 0x^3$$

$$f(x) = -2x + \frac{1}{2}x^2 - 2 = (-2) + (-2)x + \frac{1}{2}x^2 + (\frac{-11}{3})x^3$$

$$f(x^2) = -\frac{1}{3}x^3 + 4x - \frac{8}{3} = (\frac{-8}{3}) + 4x + 0x^2 + (\frac{-11}{3})x^3$$

Hence

$$A := [f]_{S_1}^{S_2} = \begin{pmatrix} -2 & -2 & \frac{-8}{3} \\ 1 & -2 & 4 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{-11}{3} \end{pmatrix}.$$

Since

$$\ker(f) \cong \ker([f]_{S_1}^{S_2}),$$

and the above matrix has trivial nullspace,

$$\ker(A) = \ker([f]_{S_1}^{S_2}) = \{\mathbf{0}\} = \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}$$

we conclude that

$$\ker(f) = \{0 \text{ (zero polynomial)}\},$$

where 0 denotes the zero polynomial.
Therefore, f is injective.
(c)
By the Rank–Nullity Theorem,

$$\text{rank}(f) + \text{nullity}(f) = \dim(\mathbb{R}[x]_{\leq 2}).$$

Since $\ker(f) = \{0\}$, we have $\text{nullity}(f) = 0$. $\implies \text{rank}(f) = 3$, however, $\dim(\mathbb{R}[x]_{\leq 3}) = 4$

$$\implies \text{rank}(f) \neq \dim\left(\underbrace{\mathbb{R}[x]_{\leq 3}}_{\text{co-domain}}\right)$$

$\implies$ f is NOT surjective. $\square$

3 Composition of linear maps and matrix multiplication

3.1 Definition

Definition 5.3.1. Let $f : A \to B$ and $g : B \to C$ be two functions. The *composition* of f and g is the function $g \circ f : A \to C$ is defined by

$$(g \circ f)(a) = g(f(a)), \qquad \forall\, a \in A.$$

For $f : A \to A$ then we denote $f^n := \underbrace{f \circ f \circ \cdots \circ f}_{n \text{ times}}$.

3.2 Theorem

Theorem 5.3.1. Let $f : V \to U$ and $g : U \to W$ be linear maps between finite-dimensional vector spaces over a field F. Fix

$$\text{a basis } B_1 \text{ of } V \text{ , a basis } B_2 \text{ of } U \text{ , a basis } B_3 \text{ of } W$$

Then we have an equality of matrices:

$$[g \circ f]_{B_1}^{B_3} = [g]_{B_2}^{B_3} \cdot [f]_{B_1}^{B_2}.$$

Proof. Let

$$\begin{aligned} B_1 &= \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m\} \text{ be a basis of } V \\ B_2 &= \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n\} \text{ be a basis of } U \\ B_3 &= \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_p\} \text{ be a basis of } W \end{aligned}$$

Let

$$\begin{aligned} [f]_{B_1}^{B_2} &= (\alpha_{ij})_{i,j} \text{ which means } f(\mathbf{v}_j) = \sum_{k=1}^{n} \alpha_{kj} \mathbf{u}_k \\ [g]_{B_2}^{B_3} &= (\beta_{ij})_{i,j} \text{ which means } g(\mathbf{u}_k) = \sum_{i=1}^{p} \beta_{ik} \mathbf{w}_i \end{aligned}$$

We now determine the matrix

$$[g \circ f]_{B_1}^{B_3}.$$

For each basis vector $\mathbf{v}_j$,

$$
\begin{aligned}
(g\circ f)(\mathbf{v}_j) &= g(f(\mathbf{v}_j)) \\
&= g\left(\sum_{k=1}^{n}\alpha_{kj}\mathbf{u}_k\right) \\
&= \sum_{k=1}^{n}\alpha_{kj}g(\mathbf{u}_k) \qquad \text{(since } g \text{ is linear)} \\
&= \sum_{k=1}^{n}\alpha_{kj}\sum_{i=1}^{p}\beta_{ik}\mathbf{w}_i \\
&= \sum_{i=1}^{p}\left(\sum_{k=1}^{n}\beta_{ik}\alpha_{kj}\right)\mathbf{w}_i.
\end{aligned}
$$

Hence, the (i,j)-entry of $[g\circ f]_{B_1}^{B_3}$ is $\sum_{k=1}^{n}\beta_{ik}\alpha_{kj}$, which is exactly the (i,j)-entry of the matrix product $[g]_{B_2}^{B_3}\cdot[f]_{B_1}^{B_2}$.
Therefore,

$$[g\circ f]_{B_1}^{B_3} = [g]_{B_2}^{B_3}\cdot[f]_{B_1}^{B_2}.$$

□

3.3 Example

3.3.1 Composition of Matrix Multiplication

The point (a,b) in $\mathbb{R}^2$ is first rotatedcounterclockwise by $\frac{\pi}{4}$ about the origin and then reflected about the line $y=\sqrt{3}\,x$. Find the resulting point.

Solution. Let

$$
\begin{aligned}
B &\text{ be the standard basis of } \mathbb{R}^2 \\
T_1 &: \text{rotation by } \frac{\pi}{4} \text{ (anti-clockwise)} \\
T_2 &: \text{reflection along the line } y=\tan\left(\frac{\pi}{3}\right)x
\end{aligned}
$$

Then

$$[T_1]_B^B = \begin{pmatrix}\cos\frac{\pi}{4} & -\sin\frac{\pi}{4}\\ \sin\frac{\pi}{4} & \cos\frac{\pi}{4}\end{pmatrix} = \begin{pmatrix}\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}.$$

$$[T_2]_B^B = \begin{pmatrix}\cos\frac{2\pi}{3} & \sin\frac{2\pi}{3}\\ \sin\frac{2\pi}{3} & -\cos\frac{2\pi}{3}\end{pmatrix} = \begin{pmatrix}-\frac{1}{2} & \frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & \frac{1}{2}\end{pmatrix}.$$

Hence,

$$\begin{aligned}
[T_2 \circ T_1]_B^B &= [T_2]_B^B \cdot [T_1]_B^B \text{ (matrix multiplication)} \\
&= \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \\
&= \begin{pmatrix} \frac{-1+\sqrt{3}}{2\sqrt{2}} & \frac{1+\sqrt{3}}{2\sqrt{2}} \\ \frac{1+\sqrt{3}}{2\sqrt{2}} & \frac{1-\sqrt{3}}{2\sqrt{2}} \end{pmatrix}.
\end{aligned}$$

Therefore,

$$\begin{pmatrix} a \\ b \end{pmatrix} \longmapsto \begin{pmatrix} \frac{-1+\sqrt{3}}{2\sqrt{2}} & \frac{1+\sqrt{3}}{2\sqrt{2}} \\ \frac{1+\sqrt{3}}{2\sqrt{2}} & \frac{1-\sqrt{3}}{2\sqrt{2}} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{-1+\sqrt{3}}{2\sqrt{2}}a + \frac{1+\sqrt{3}}{2\sqrt{2}}b \\ \frac{1+\sqrt{3}}{2\sqrt{2}}a + \frac{1-\sqrt{3}}{2\sqrt{2}}b \end{pmatrix}}_{\text{in standard basis of B}}.$$

$\square$

4 Inverse of a matrix

4.1 Definition

Definition 5.4.1. Let $A \in M_n(F)$ be a square matrix. If there exists a matrix $B \in M_n(F)$ s.t. $AB = BA = I_n$, then

1. we say that A is *invertible* (or *non-singular*); and
2. we call B the *inverse* of A and denote it by A^{-1}.

4.2 Theorem

Theorem 5.4.1. Let $A \in M_{m\times n}(F)$. Suppose there exists $B \in M_{n\times m}(F)$ s.t.

$$BA = I_n \text{ and } AB = I_m.$$

Then $n = m = \operatorname{rank}(A)$.

Proof. We shall show that $\operatorname{rank}(A) = m = n$.
(1) First observe that

$$\operatorname{rank}(A) = \operatorname{rank}(A\overbrace{BA}^{I_n}) \leq \operatorname{rank}(AB) = \operatorname{rank}(I_m) = m,$$

On the other hand,

$$m = \operatorname{rank}(I_m) = \operatorname{rank}(AB) \leq \operatorname{rank}(A).$$

Therefore,

$$\operatorname{rank}(A) = m.$$

(2) Similarly,

$$\operatorname{rank}(A) = \operatorname{rank}(ABA) \leq \operatorname{rank}(BA) = \operatorname{rank}(I_n) = n,$$

Also,

$$n = \operatorname{rank}(I_n) = \operatorname{rank}(BA) \leq \operatorname{rank}(A).$$

Hence,

$$\operatorname{rank}(A) = n.$$

Therefore, if A has an inverse, then

$$n = m = \operatorname{rank}(A).$$

Remark. Rank inequalities

$$\begin{cases} \operatorname{rank}(f \circ g) \leq \operatorname{rank}(f) \\ \operatorname{rank}(f \circ g) \leq \operatorname{rank}(g) \end{cases}$$

and matrix version

$$\begin{cases} \operatorname{rank}(AB) \leq \operatorname{rank}(A) \\ \operatorname{rank}(AB) \leq \operatorname{rank}(B) \end{cases}$$

□

□

Theorem 5.4.2. Let $f : V \to W$ be an isomorphism of finite-dimensional vector spaces. Fix any basis B_1 of V and B_2 of W. Then we have an equality of matrices

$$\left([f]_{B_1}^{B_2}\right)^{-1} = [f^{-1}]_{B_2}^{B_1}.$$

In other words, the inverse of a matrix represents the inverse of a linear map.

Proof. By Theorem 5.3.1,

$$[f^{-1}]_{B_2}^{B_1}[f]_{B_1}^{B_2} = [f^{-1} \circ f]_{B_1}^{B_1}.$$

Since $f^{-1} \circ f = \mathrm{id}_V$, we obtain

$$[f^{-1} \circ f]_{B_1}^{B_1} = [\mathrm{id}_V]_{B_1}^{B_1} = I_n, \text{ where } n = \dim(V)$$

Therefore,

$$[f^{-1}]_{B_2}^{B_1} = \left([f]_{B_1}^{B_2}\right)^{-1}.$$

□

5 Remark

Vector space / Linear map world	**Vector / Matrix world**
$f : V \to W$	$[f]_{B_1}^{B_2}$ (matrix)
Composition $f \circ g$	Multiplication $[f \circ g]_{B_1}^{B_3} = [f]_{B_2}^{B_3}[g]_{B_1}^{B_2}$
Addition $f + g$, Scaling λf	Addition $[f + g]_{B_1}^{B_2} = [f]_{B_1}^{B_2} + [g]_{B_1}^{B_2}$, Scaling $[\lambda f]_{B_1}^{B_2} = \lambda [f]_{B_1}^{B_2}$
Inverse $f^{-1} : W \to V$	Matrix inverse $[f^{-1}]_{B_2}^{B_1} = ([f]_{B_1}^{B_2})^{-1}$

Question: When is a matrix invertible?